Aaradhya Dev Tamrakar

Applied Machine Learning · MLOps · Computer Vision · Agentic AI

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Final-year engineering undergraduate and Fusemachines AI Fellow building applied ML pipelines, vision evaluation frameworks, and production agentic services. Engineered a 5-stage Text-to-SQL engine with 100% benchmark execution accuracy, demographic fairness auditing over 97K images, and full-stack microservices using FastAPI, Docker, and PostgreSQL. Experienced in PyTorch, scikit-learn, XGBoost, and MLOps.

EDUCATION

Bachelor of Engineering (B.E.) in Electronics, Communication and Information Engineering | December 2022 – Expected January 2027

Kathmandu Engineering College, Institute of Engineering (IOE), Tribhuvan University • Cumulative Average: ~75% across completed semesters

Relevant Coursework: Machine Learning · Signals and Systems · Digital Electronics · Control Systems · Communication Systems · Engineering Mathematics · Probability and Statistics · Object-Oriented Programming (C++) · Computer Networks · Microprocessors

FELLOWSHIPS & APPLIED ML EXPERIENCE

Fuse AI Fellow — Fusemachines AI Fellowship 2026 | 2026 – Present (Active)

- Engineered a five-stage agentic Text-to-SQL pipeline (Planner → Generator → Validator → Executor → Summarizer), achieving **100% execution success and answer accuracy** on a 50-question project benchmark.
- Built containerized AI microservices and evaluation APIs using **FastAPI**, **PostgreSQL**, and **Docker**.
- Developed customer churn and lifetime value (CLV) prediction pipelines (**5-fold CV ROC-AUC 0.841**) utilizing **SMOTE**, cost-sensitive threshold tuning, and **SHAP** explainability.
- Conducted Bayesian estimation and Gaussian-process regression with **PyMC/ArviZ**, and performed unsupervised segmentation (**K-Means**, **DBSCAN**) on **~500K retail transaction records**.
- Assembled a four-model time-series forecasting ensemble achieving **MASE 2.44** on the UCI Online Retail II dataset.

NSSR DataCamp Fellow (Cohort 2) — Nepalese Society of Student Researchers | April 2026 – Present

- Selected in competitive cohort for sponsored DataCamp access; developed six applied GenAI solutions automating data cleaning, schema normalization, and null filtering.
- Completed advanced **SQL Associate** and **Python Data Associate** coursework focused on PostgreSQL, Pandas, and inferential statistics.

RESEARCH & TECHNICAL PROJECTS

BiasAperture — Computer Vision Demographic Fairness & Bias Audit | 2026 – Present

PyTorch · Python · AIF360 · Fairlearn · FairFace · SHAP · Statistical Testing

- Co-developed an automated fairness-auditing framework with dual **AIF360** and **Fairlearn** backends computing **Disparate Impact Ratio (DIR)**, **Demographic Parity Difference (DPD)**, **Equal Opportunity Difference (EOD)**, and **Equalized Odds**.
- Audited classification performance across **126 granular race × age × gender groups** on the **FairFace benchmark (97,698 images)**, discovering significant disparity drop-offs across intersectional distribution shifts.
- Implemented rigorous statistical inference: **BCa bootstrap confidence intervals (B ≥ 1,000)**, **χ^2 and Fisher's exact tests**, **n ≥ 30 sample-size safeguards**, and **Shapley feature attribution (SHAP)** for flagged discrepancies.
- Architected clean CLI, HTML/PDF reporting pipelines, and modular test harnesses with **78/78 passing repository tests**; co-authored with Tisha Manandhar.

SPARK — Safety-Critical Wearable Edge-AI Fall Detection System | 2025 – Present

ESP32-S3 · C/C++ · TFLite Micro · PyTorch · Python · BLE · FastAPI · SHAP

- Technical lead for ML modeling, quantization, and gateway architecture in a four-person team; designed a two-tier fall detection system eliminating cloud latency and privacy hazards.
- Trained 1D-CNN on the **SisFall benchmark (38 subjects, 38,420 windows)**; achieved **0.9185 AUC-ROC (95% CI: [0.9065, 0.9305])** and **87.81% sensitivity** under subject-stratified 5-fold cross-validation with zero data leakage.
- Executed post-training integer quantization (PTQ), compressing model flash memory footprint to **18.5 KB** for deterministic, **sub-100ms** on-device confirmation on the **ESP32-S3** microcontroller.

Fleet-Orchestrator — Multi-Account Autonomous Agent Swarm & Consensus Engine | 2026 – Present

Python 3.12+ · FastAPI · SQLite WAL · asyncio · MCP · CDP · Copilot CLI

- Implemented multi-provider adapters (Copilot CLI, Gemini, Claude, local SLMs), quota-aware round-robin scheduling, and adversarial lease-stealing tests.

Super-NLM — Multi-Account NotebookLM Aggregator & MCP Hub | 2026

Python · FastAPI · React · MCP · asyncio

Edge AI Stability Detection System | 2025

Python · scikit-learn · FastAPI · Joblib · IMU Data

- Trained a Random Forest classifier predicting platform stability from simulated 6-axis IMU streams, reaching **99.8% test accuracy**.
- Serialized the pipeline via Joblib and deployed an ultra-low-latency **FastAPI REST inference endpoint** for robotics integration.

nepali-ocr-ai — Low-Resource Document Intelligence | 2026

Python 3.11 · Tesseract · Devanagari OCR · OpenXML · FastMCP

- Prototyped a hybrid Nepali document intelligence pipeline combining vision LLM OCR with local Tesseract fallback, Devanagari normalization, and Preeti ↔ Unicode transcoding.

Nexus — Personal AI Workspace & Knowledge Retrieval Hub | 2025 – 2026

FastAPI · React (Vite) · SQLite FTS5 · asyncio · TailwindCSS

LEADERSHIP & SERVICE

Vice Chair (2026 – Present); Vice Secretary (2025 – 2026) — IEEE KEC KTM Student Branch | 2025 – Present

- Direct branch strategy, executive committee operations, technical workshops, and peer learning initiatives for engineering students.
- Previously managed operational documentation and workshop logistics across department-level technical events as Vice Secretary.

Event Manager (2026 – Present); Resource Manager (2024 – 2026) — Electronics Project Club (EPC), Kathmandu Engineering College | 2024 – Present

- Plan and execute technical hackathons, project exhibitions, and hardware workshops; manage lab equipment allocation across engineering departments.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, SQL (PostgreSQL, SQLite), Kotlin, C# (.NET), VHDL, PowerShell, Bash, JavaScript / TypeScript

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, PyMC, ArviZ, SHAP, Fairlearn, AIF360, Hugging Face Transformers, OpenCV, MediaPipe

AI Systems, Backend & MLOps: FastAPI, Docker, Model Context Protocol (MCP), asyncio, MLflow, Evidently AI, Airflow, PostgreSQL, SQLite FTS5, REST APIs

Engineering Practice & Tools: Git, GitHub, Linux (Ubuntu), Jupyter, Google Colab, LaTeX / Overleaf, WordCraft, Pytest, CMake / CTest, Automated CI/CD